SeedingMinds

Assessment Name

Java_S3_030126

Module: Coding

Question 1:

Marks: +10 -0

Shopkeeper's Missing Inventory

A shopkeeper has an inventory of goods represented by an array of positive integers, where each integer represents the number of items in stock for a particular product. Some of the items are misplaced, and he wants to rearrange the inventory such that all zeroes (representing out-of-stock products) are moved to the end of the array while maintaining the order of the non-zero items.Your task is to help the shopkeeper determine the maximum possible sum of any contiguous subarray of non-zero items. The array must be modified in place to move all zeros to the end, then the maximum sum of any contiguous subarray of the remaining non-zero elemen should be calculated and returned and maximum sum of the non-zero subarray.

Constraints:

- The array should be modified in place.
- The input array will have at least 1 element and at most 100,000 elements.
- No additional arrays should be used.

Example:

Example 1-
Input: 4 0 2 0 3 7
Output: [4, 2, 3, 7, 0, 0], 16

Example 2-
Input: 7 0 4 0 3 2
Output: [7, 4, 3, 2, 0, 0],16

Explanation:

In this problem, you need to rearrange the shopkeeper's inventory by moving all zeroes (out-of-stock items) to the end of the array while keeping the order of non-zero items intact. This ensures that the merchant can easily see available stock first. The solution requires modifying the array in place without using additional storage, which is efficient and suitable for large inventories. Along with the maximum sum of the non-zero subarray.

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	`6 0 5 0 8`	`[6, 5, 8, 0, 0], 19`
2	`0 1 2`	`[1, 2, 0], 3`

Question 2:

Marks: +10 -0

Weather Station Broadcasts

In a remote region with unpredictable weather, several weather stations are responsible for broadcasting real-time data at fixed intervals. However, the intervals at which these stations broadcast their data differ, causing synchronization issues. To optimize the system, you are tasked with finding the minimum time interval at which all stations can broadcast together. The system needs to calculate the synchronized broadcast interval and adjust the intervals for each station accordingly. Your task is to write a program th takes the minimum broadcast interval and the broadcast intervals of each weather station, then finds the minimum synchronized interval and determines how much each station needs to adjust its interval to match the synchronized time.

Constraints:

- The number of weather stations is not fixed, but each station has a distinct broadcast interval.
- The minimum interval is always a positive integer and must be determined based on the greatest common divisor (GCD) of the given intervals.
- Each station's interval is a positive integer, ranging from 1 to 10^6

Example:

Example 1:

Input: 15
12 18 24
Output: 72
60 54 48

Example 2:

Input: 40
8 16
Output: 40
32 24

Explanation:

In this task, you are responsible for synchronizing weather stations across a vast region. Each station broadcasts weather data at different time intervals, but to improve efficiency, all stations need to broadcast their data simultaneously at the same interval. Your j is to create a system that determines the minimum possible interval at which all weather stations can broadcast together. Additionall for each station, calculate the necessary adjustment to their original broadcast intervals so that they match the synchronized broadcast interval.

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	18 3 6 9	18 15 12 9
2	30 15 20 25 30	300 285 280 275 270

Question 3:

Marks: +10 -0

Q4 - Hollow Patterns

Write a Program to Print the following Pattern?

If Input is 5 then Print

```
*
* *
*   *
*     *
* * * * *
```

Constraints:

Input :- First Line of Input Consists of One Integer Value.

Output :- Print the Given Pattern.

Constraints :- Given Input Must be Greater than Zero or Else Print **"Invalid Input"**.

Example:

Input 1 : 5

Output 1:

```
*
* *
*   *
*     *
* * * * *
```

Input 2 : 3

Output 2:

```
*
* *
* * *
```

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT

1	5	`*`
		`* *`
		`*  *`
		`*  *`
		`* * * * *`

2	3	`*`
		`* *`
		`* * *`

Question 4:

Marks: +10 -0

Q4 - Methods - Alternative odd numbers between Given Numbers

Write a program to print the Alternative odd numbers by using Method between the Given Numbers ?

Please make sure that code must be in this format

Do not change the code block

Constraints:

Input : First Line consists of **Input 1**(integer n1).

Second Line consists of **Input 2**(integer n2).

Output : Print the alternative numbers between the given numbers.

Constraints : If the given n1 and n2 value is lessthan convert them to **Positive values**.

If there are no odd numbers between the given n1 and n2, Print **"No Odd Numbers Between them"**

If the given n1 value is only equal to zero, Print **"First Input is Zero"**.

If the given n2 value is only equal to zero, Print **"Second Input is Zero"**.

If both the n1 and n2 values are equal to zero, Print **"Both Inputs are Zero"**.

Example:

Input 1 : 1 19

Output 1 : 3, 7, 11, 15.

Input 2 : 0 0

Output 2 : Both Inputs are Zero

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	1 19	3, 7, 11, 15.
2	0 0	Both Inputs are Zero